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In space, astronauts are exposed to a complex and unique radiation environment composed of trapped proton and electron radiation, galactic cosmic rays (GCR; energetic protons and heavy charged ions, HZE), and solar particle events (SPE), primarily protons. Although these radiations can result in appreciable cumulative doses to astronauts during long-duration missions, the exposures occur at low fluences and fluence rates, such that particle traversals through individual cells in an astronaut's body may be well separated in tissue location and time. Under such conditions, so-called non-targeted effects such as bystander responses may have increased significance. The radiation-induced bystander effect is the occurrence of biological responses in unirradiated cells near to or sharing medium with cells actually traversed by radiation. We hypothesize that at the low fluences of protons and HZE particles to which astronauts are exposed in space, a significant fraction of the cellular changes may be due to bystander responses, in addition to the damages caused in cells directly traversed by ionizing particles or their associated delta rays. We have the

Task Description:	following specific aims. 1. Test the hypothesis that bystander effects occur at particle fluences so low that only 1-2 % of cells in a culture dish are traversed by an energetic proton or HZE particle. 2. Extend our low fluence work to studies using repeated exposures of cells to low fluences and mixed (actually sequential) beams of more than one particle type. This will test the hypothesis that cells that have survived an initial traversal and sent a bystander signal, can again send a signal when traversed once again by a particle or can subsequently respond as bystanders to a hit in a neighboring cell. 3. Our studies have indicated that reactive oxygen species (ROS) are at least partially involved in transmittal of the Fe ion-induced bystander signal from hit to unirradiated cells. We will test the hypothesis that non-nuclear damage-initiated, metabolically produced ROS in hit cells are involved in the early steps of bystander signaling. Additionally, we will investigate chemical countermeasures to reduce the bystander responses based on removal of ROS or prevention of their formation. 4. In tissues, cells do not exist in isolation, and there is increasing interest in how signaling between cell types (especially stromal and epithelial) influences cellular responses to stresses. We will test the hypothesis that radiation-induced bystander signals can be sent between differing cell types, but the characteristics of the responses depend on the cell types and the types of radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although our studies are specifically geared to providing information on the possible biological effects, especially cancer induction, of low fluences of energetic particles on astronauts, the data may have relevance to cancer therapy and the effects of low doses of radiation on radiation workers (e.g., those involved in clean-up of radioactive sites) and the general public. With regards to cancer, there is increasing use of protons (energies of 65-250 MeV) as an effective treatment modality in radiation oncology because they provide a great therapeutic advantage in some types of cancers due to exquisite dose localization. Over 40,000 patients have been treated world-wide with protons, and several new proton centers have opened in the last few years or are under construction in the US. However, despite the clinical success of protons, some concerns exist regarding the possible side-effects, including the possibility of cancer caused by the low doses received by normal tissues. Additionally, there is increasing interest in the use of carbon ions in cancer therapy, with carbon ion treatment centers now in use in Japan and Germany, and the potential being discussed in the US. Our studies could provide important and useful new insight on the effects of these energetic particles on biological systems and the mechanisms by which they act. Such findings would be expected to translate into helping to develop effective ways to reduce possible side-effects, including cancer, from the treatments.
	Ion-induced media-mediated bystander effects. Previously we reported our use of a transwell insert culture method to demonstrate media-mediated bystander responses, using a number of different endpoints, in AG01522 normal human fibroblasts after low LET X-irradiation (1) and 1 GeV/n iron ions (2), and, more recently, 1 GeV protons, the ion work done using the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory. The work showed that with increasing radiation dose, micronuclei (MN) formation increases in irradiated cells, with the Fe ions being significantly more efficient at producing MN than X-rays or protons. However, in unirradiated bystander cells sharing medium with irradiated cells, MN formation increases about 2-fold at the lowest radiation dose used (10 cGy), then remains constant to the highest dose, irrespective of whether iron ions or protons were used. Similar dose response data for irradiated and bystander cells were obtained using the endpoints of formation of nuclear foci of DNA repair-related proteins, gamma-H2AX or 53BP1, as markers of DNA double strand breaks, using doses of 5 cGy and above. Although at doses above about 10 cGy the magnitude of the bystander effects is small compared to the damage from direct irradiation, at lower doses the bystander responses make a significant contribution to the overall cell damage measured. Bystander effects induced at very low particle fluences (aim 1). In the previous reporting period, we presented preliminary data demonstrating media-mediated bystander effects after iron ion exposures at fluences as low as 0.02 particles/cell (about 2% of cells in dish irradiated; roughly a dose of 0.45 mGy of iron ions) for the endpoint of MN and down to 0.008 particles/cell for 53BP1 foci formation. In contrast, with protons, the very early data suggested that bystander responses were not seen until fluences of 0.2 or 2 protons/cell with the MN and 53BP1 foci endpoints, respectively. This year we have repeated these studies and extended to lower fluences with both endpoints and both 1 GeV/n Fe and protons. Although additional repeats are needed to reduce the size of the error bars, with the endpoint of MN formation, some general trends are apparent. First, in directly irradiated cells, much higher fluences are needed to see MN with both ions than are needed for expression of bystander effects. At this time we do not understand why responses should be seen at lower doses in the bystander population than in the irradiated population since, at these low fluences, most of the cells in the irradiated population are not traversed by a particle and hence actually are “bystanders”. Second, although our very early data suggested that bystander responses were elicited at much lower fluences of iron ions than of protons, this does not appear to be holding up with replicate experiments. Data showing similar trends have been obtained also with 53BP1 foci formation as the endpoint. The results just presented were all obtained using exponentially growing cells. As we pointed out last year, much of the pioneering work by others in bystander responses was done using confluent cells [e.g., (3-5)], and most cells in the human body are not exponentially growing, so we were interested in investigating whether results similar to those just described are obtained with confluent cells. This is important from a mechanistic standpoint, since it has been proposed that gap junctions are important for the bystander effect in confluent cells, but gap junctions are not relevant in studies done using our insert system. The data we described last year were quite preliminary, and the results of fuller studies have now been obtained. As shown previously, exponential cells can send and respond to media-mediated bystander signals. Our new data with confluent cells, however, clearly indicate that in this system confluent AG01522 cells irradiated with 1 GeV/n iron ions, 1 GeV protons or 250 kVp X-rays are not able to send media-mediated bystander signals to either confluent or exponential cells to induce formation of MN. Unfortunately, with MN as the endpoint there is a technical difficulty using confluent cells as the bystander signal-receiving cells, because cells must be dividing to express MN. We attempted studies in which we left irradiated exponential cells with bystander confluent cells for different times (1-24 h) then replated the confluent cells at low cell density to allow cell division. In those studies we did not see any increase in MN in the replated bystander cells. Since we were concerned that the timing of replating or

Task Progress:

the membrane alterations that are involved in trypsinizing and replating cells might have interfered with expression of the bystander response, we have more recently tried the endpoint of 53BP1 foci formation, a response that does not require cells to divide and is seen in irradiated confluent cells. As shown in preliminary data with X-rays, under these conditions there is an increase in damage in confluent bystanders when exponential cells are irradiated; in other words, confluent cells can respond to a bystander signal. We are now doing this last experiment with iron ions and protons to ascertain whether the same response as with X-rays will be obtained.

Bystander responses in “mixed” radiation fields (aim 2).

Starting during the last part of the funding period of our previous NASA grant and continuing in aim 2 of this current grant, we have been among the first groups at NSRL to use the “mixed”, actually sequential, beams of Fe ions and protons. These studies were designed to assess possible interactions between the two types of radiation in cells, when the irradiations were separated in time. In particular, we asked whether adaptive responses, in which a small “conditioning” dose (e.g., from protons) could decrease cell response to a subsequent larger dose (e.g., from iron ions) in the cells traversed by radiation, as well as in cells that are bystanders to the “hit” cells. Our preliminary interpretations of the initial data are as follows. First, when “bystander” cells were irradiated with 1 cGy protons, then introduced into co-culture with iron-irradiated cells either immediately, 30 min or 3 h later, no bystander effect was observed, i.e., the % binucleate cells with MN in the bystander population was the same as in untreated control populations. This suggests that the cells became “adapted” against the iron ion-induced bystander signal by first receiving a low dose of protons. Second, the typical bystander response (i.e., increased MN) was restored when the time delay between the initial proton irradiation of “bystander” cells and the iron ion exposure to the “hit” cells and co-culture was extended to 24 h, suggesting the “adaptation” against the bystander response was lost in 24 h. Third, when the signaling (“hit”) cells were exposed to 1 cGy protons at different times (0-24 h) prior to iron-irradiation, the bystander effect in non-irradiated cells sharing medium with them was the same as when the hit cells received only one dose. Together, these three observations suggest that signal-receiving cells can be adapted against a bystander signal by a prior low dose of radiation, although the adaptation decays by 24 h. On the other hand, the ability of signaling cells to generate the signal cannot be adapted against.

Since reactive oxygen species (ROS) are believed to be involved in bystander signaling (see discussion of aim 3 in next paragraph), we hypothesized that pretreatment of bystander cells with a low, non-toxic dose of hydrogen peroxide could alter bystander responses induced by radiation. Our data to test this showed that pretreatment of bystander cells with 5 ?M H2O2 did not alter expression of the “normal” bystander response in the cells sharing medium with cells irradiated with iron ions or X-rays, unlike the “adaptive” response elicited by pre-irradiating bystander cells with a low dose of protons. This early finding suggests that although ROS are involved in signaling from irradiated to unirradiated bystander cells, oxidative stress may not be a factor in the pathway for development of the adaptive response. Clearly, further studies are needed to address the underlying mechanisms.

Role of ROS in bystander signaling (aim 3).

We have shown that SOD, catalase, and c-PTIO, which remove superoxide, hydrogen peroxide and nitric oxide, respectively, when added to medium reduce the Fe ion-induced media-mediated bystander effects (as well as X-ray-induced bystander effects), indicating a role of reactive oxygen species (ROS) in ion-induced bystander effects (2). We have continued our mechanistic studies of the role of ROS in signaling the bystander response (aim 3). We now have confirmed our findings reported initially last year that pre-treating irradiated cells with DPI, an inhibitor of the membrane-bound superoxide-generating enzyme NADPH oxidase, reduces bystander effects after both iron ion and proton irradiation. The DPI has no effect on response of irradiated cells, suggesting the source of the superoxide involved in media-mediated bystander signaling may, at least in part, be NADPH oxidase. This is an important finding because it indicates that non-DNA damage, i.e., membrane-located, processes may be important in the signaling in the bystander response.

Bystander signaling in mixed cell types (aim 4).

Under the auspices of our first NASA grant (# NAG2-1642), we had obtained preliminary data on signaling between the “mixed” cell types of AG01522 human fibroblasts and hTERT-immortalized human keratinocytes (6) that was presented in the application for this grant. The results had suggested that with 250 kVp X-rays, and using the transwell insert system, bystander signaling occurs from irradiated fibroblasts to unirradiated keratinocytes and vice versa. However, with 1GeV/n iron ions, our preliminary data suggested that the bystander response in keratinocytes co-cultured with irradiated fibroblasts is delayed, and no response is seen in the reverse direction. We have now confirmed the previous findings with iron ions and are extending the mixed culture studies to irradiation with 1 GeV protons and 290 MeV/n carbon ions to determine if there is an LET dependence for the lack of signaling from keratinocytes to fibroblasts. If LET is important, we would predict that the low LET protons should give the same results as X-rays did, and carbon ions, of intermediate LET, might be expected to give an intermediate response. This is an important study, as it may shed light on some conditions for a radiation quality dependence for the bystander effect.

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Bibliography Type:	Description: (Last Updated: 11/13/2018)
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Abstracts for Journals and Proceedings	Yang H, Held KD. "Bystander responses in AG01522 normal human skin fibroblasts at very low fluences of iron ions or energetic protons." Presented at the 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 2008. Proceedings of the 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 2008. , Sep-2008
Abstracts for Journals and Proceedings	Yang H, Magpayo N, Held KD. "Bystander signaling in exponential and confluent human fibroblasts at very low fluences of iron ions or energetic protons." Poster presentation at the NASA Human Research Program Investigators' Workshop, League City, TX, February 2009. Proceedings of the NASA Human Research Program Investigators' Workshop, February 2009. , Feb-2009
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